MAT 2040 Linear Algebra

Assignment 4

- Released date: 2024/06/24.
- Due: 2024/06/27.
- Late submission is NOT acceptable.
- Please submit your assignment as a PDF file titled "student ID + HW4".

Question 1. Determine whether the following sets are bases for $\mathbb{R}^3$.

(a)
$$\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}1\\1\\0\end{bmatrix}, \begin{bmatrix}1\\1\\1\end{bmatrix}$$

(b)
$$\begin{bmatrix}1\\0\\1\end{bmatrix}, \begin{bmatrix}0\\0\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}$$

(c)
$$\begin{bmatrix}1\\0\\-2\end{bmatrix}, \begin{bmatrix}3\\2\\-4\end{bmatrix}, \begin{bmatrix}-3\\-5\\1\end{bmatrix}$$

Solution

(a) We can use the Gaussian elimination method to check for linear independence. Let's form an augmented matrix with these vectors as columns and a

zero vector as the last column:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right]$$

This matrix is already in row echelon form. We can see that there are no rows with all zeros in the left side, so the rank of the matrix is 3, which is the same as the dimension of $\mathbb{R}^3$. This means the given vectors are linearly independent. .

(b) Because of the existence of 0 vector, these 3 vectors are not linearly independent, so this set of vectors is not a basis for $\mathbb{R}^3$.

(c)

Let $A = \begin{bmatrix} 1 & 3 & -3 \\ 0 & 2 & -5 \\ -2 & -4 & 1 \end{bmatrix}$. We can perform Gaussian Elimination on the matrix to obtain the matrix B: $\begin{bmatrix} 1 & 3 & -3 \\ 0 & 2 & -5 \\ 0 & 0 & 0 \end{bmatrix}$

The matrix B only has 2 pivot positions, so the set doesn't form a basis for $\mathbb{R}^3$.

Question 2. Find a basis for the solution set of the following system:

$$\begin{bmatrix} -2 & 4 & -2 & -4 \\ 2 & -6 & -3 & 1 \\ -3 & 8 & 2 & -3 \end{bmatrix} \mathbf{x} = \mathbf{0}$$

Solution

Let $A = \begin{bmatrix} -2 & 4 & -2 & -4 \\ 2 & -6 & -3 & 1 \\ -3 & 8 & 2 & -3 \end{bmatrix}$, we can do the elementary row operations to get the echolon form.

$$\begin{bmatrix} -2 & 4 & -2 & -4 \\ 2 & -6 & -3 & 1 \\ -3 & 8 & 2 & -3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 & 1 & 2 \\ 2 & -6 & -3 & 1 \\ -3 & 8 & 2 & -3 \end{bmatrix} \rightarrow$$

$$\begin{bmatrix} 1 & -2 & 1 & 2 \\ 0 & -2 & -5 & -3 \\ 0 & 2 & 5 & 3 \end{bmatrix} \to \begin{bmatrix} 1 & -2 & 1 & 2 \\ 0 & 2 & 5 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 6 & 5 \\ 0 & 1 & 5/2 & 3/2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

So from the last matrix, we know that x_3 and x_4 are free variables, and $x_1 = -6x_3 - 5x_4$, $x_2 = -5/2x_3 - 3/2x_4$.

So we can write the solution:

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -6 \\ -5/2 \\ 1 \\ 0 \end{bmatrix} x_3 + \begin{bmatrix} -5 \\ -3/2 \\ 0 \\ 1 \end{bmatrix} x_4$$

So a basis is: $\begin{bmatrix} -6 \\ -5/2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -5 \\ -3/2 \\ 0 \\ 1 \end{bmatrix}$

Question 3. In each of the following, find the dimension of the subspace of P_3 spanned by the given vectors:

(a) $x, x-1, x^2+1$

(b) $x, x-1, x^2+1, x^2-1$

Solution

(a) We should find the coefficients c_1, c_2, c_3 such that:

$$\begin{aligned} &c_1x + c_2(x-1) + c_3(x^2+1) = 0, \textit{i.e.}, \\ &(c_3 - c_2) + (c_1 + c_2)x + c_3x^2 = 0 \cdot 1 + 0 \cdot x + 0 \cdot x^2 \end{aligned}.$$

So we can have:

$$\begin{cases} -c_2 + c_3 = 0 \\ c_1 + c_2 = 0 \\ c_3 = 0 \end{cases}$$

Therefore, $c_1 = c_2 = c_3 = 0$. So the polyomials in (a) are linearly independent. The dimension is **3**.

(b)

$$\begin{aligned} &c_1x + c_2(x-1) + c_3(x^2+1) + c_4(x^2-1) = 0 \\ &(c_3 - c_2 - c_4) + (c_1 + c_2)x + (c_3 + c_4)x^2 = 0 \cdot 1 + 0 \cdot x + 0 \cdot x^2 \end{aligned}.$$

So we have:

$$\begin{cases} -c_2 + c_3 - c_4 = 0 \\ c_1 + c_2 = 0 \\ c_3 + c_4 = 0 \end{cases}$$

So there exist $c_1 = -2, c_2 = 2, c_3 = 1, c_4 = -1$ that make $c_1x + c_2(x-1) + c_3(x^2+1) + c_4(x^2-1) = 0$, so they are not linearly independent. So dimensions is **3**.

Question 4. Let $\{x, 1\}$ and $\{2x-1, 2x+1\}$ be ordered bases for P_2, find the transition matrix representing the change in coordinates from $\{2x-1, 2x+1\}$ to $\{x, 1\}$.

Solution

The coordinate vector for $2x-1$ of the old basis under the new basis is:

$$[2x-1]_{\{x,1\}} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

The coordinate vector for $2x+1$ of the old basis under the new basis is:

$$[2x+1]_{\{x,1\}} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Therefore, put the two coordinates vectors together, the transition matix is:

$$\begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix}$$

Question 5. Given $\mathbf{x}_1=(1,1,1)^T$ and $\mathbf{x}_2=(3,-1,4)^T$:

(a) Do $\mathbf{x}_1$ and $\mathbf{x}_2$ span $\mathbb{R}^3$? Explain.

(b) Let $\mathbf{x}_3$ be a third vector in $\mathbb{R}_3$ and set $X = (\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3)$. What condition(s) would X have to satisfy in order for $\mathbf{x}_1$, $\mathbf{x}_2$, and $\mathbf{x}_3$ to form a basis for $\mathbb{R}_3$?

(c) Find a third vector $\mathbf{x}_3$ that will extend the set $\{\mathbf{x}_1, \mathbf{x}_2\}$ to a basis for $\mathbb{R}^3$.

Solution

(a) To check whether $\mathbf{x}_1$ and $\mathbf{x}_2$ span $\mathbb{R}^3$, we need to determine whether every vector in $\mathbb{R}^3$ can be expressed as a linear combination of $\mathbf{x}_1$ and $\mathbf{x}_2$. Let $\mathbf{x}_3=(1,0,0)^T$. Then $\mathbf{x}_3$ is a vector in $\mathbb{R}^3$ that cannot be expressed as a linear combination of $\mathbf{x}_1$ and $\mathbf{x}_2$. Therefore, $\mathbf{x}_1$ and $\mathbf{x}_2$ do not span $\mathbb{R}^3$.

(b) In order for $\mathbf{x}_1$, $\mathbf{x}_2$, and $\mathbf{x}_3$ to form a basis for $\mathbb{R}^3$, they must satisfy the

condition:

The vectors $\mathbf{x}_1$, $\mathbf{x}_2$, and $\mathbf{x}_3$ must span $\mathbb{R}^3$, which means that any vector in $\mathbb{R}^3$ can be expressed as a linear combination of $\mathbf{x}_1$, $\mathbf{x}_2$, and $\mathbf{x}_3$.

To determine the condition that X must satisfy, we can set $\mathbf{x}_3$ in X to $(a, b, c)^T$. $\mathbf{x}_1$, $\mathbf{x}_2$ and $\mathbf{x}_3$ then form the matix:

$$A = \begin{bmatrix} 1 & 3 & a \\ 1 & -1 & b \\ 1 & 4 & c \end{bmatrix}$$

The reduced row echelon form of the matrix A is:

$$B = \begin{bmatrix} 1 & 3 & a \\ 0 & 1 & \frac{a}{4} - \frac{b}{4} \\ 0 & 0 & -\frac{5a}{4} + \frac{b}{4} + c \end{bmatrix}$$

Because $\mathbf{x}_1$, $\mathbf{x}_2$ and $\mathbf{x}_3$ should be linearly independent, we have: $-\frac{5a}{4}+\frac{b}{4}+c \neq 0$ Therefore, if X meets the condition that, $-\frac{5a}{4} + \frac{b}{4} + c \neq 0$ for $\mathbf{x}_3$, then $\mathbf{x}_1$, $\mathbf{x}_2$ and $\mathbf{x}_3$ can form a basis for $\mathbb{R}^3$.

(c) Based on (b), all we need to do is to find $\mathbf{x}_3$ that is linearly independent of $\mathbf{x}_1$ and $\mathbf{x}_2$. Then one possible solution is that $\mathbf{x}_3$=$(1, 0, 0)^T$.

Question 6. The vectors $\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \mathbf{x}_2 = \begin{bmatrix} 2 \\ 5 \\ 4 \end{bmatrix}, \mathbf{x}_3 = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}, \mathbf{x}_4 = \begin{bmatrix} 2 \\ 7 \\ 4 \end{bmatrix}, \mathbf{x}_5 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ span $\mathbb{R}^3$. Pare down the set $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4, \mathbf{x}_5\}$ to form a basis for $\mathbb{R}^3$.

Solution

To pare down the set $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4, \mathbf{x}_5$ to form a basis for $\mathbb{R}^3$, we need to eliminate any redundant vectors. One way to do this is to use elementary row operations to reduce the augmented matrix:

$$\left[\begin{array}{ccccc|c} 1 & 2 & 1 & 2 & 1 & 0 \\ 2 & 5 & 3 & 7 & 1 & 0 \\ 2 & 4 & 2 & 4 & 0 & 0 \end{array}\right]$$

to obtain the row echelon form:

$$\left[\begin{array}{ccccc|c} 1 & 0 & -1 & -4 & 3 & 0 \\ 0 & 1 & 1 & 3 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{array}\right]$$

In this case, the first, second, and fifth columns of the row echelon form do not have leading ones, so the corresponding vectors $\mathbf{x}_1, \mathbf{x}_2$ and $\mathbf{x}_5$ form a basis for $\mathbb{R}^3$. Therefore, we can pare down the original set to the basis $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_5\}$.

Question 7. Let

$$\mathbf{u}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \mathbf{u}_2 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \mathbf{u}_3 = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}, \mathbf{v}_1 = \begin{bmatrix} 4 \\ 6 \\ 7 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix},$$

(a) Find the transition matrix from $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ to $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$.

(b) If $\mathbf{x} = 2\mathbf{v}_1 + 3\mathbf{v}_2 - 4\mathbf{v}_3$, determine the coordinate of $\mathbf{x}$ with respect to $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$.

Solution

(a) Let $U = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$, $V = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$. This question asks for a matrix A such that:

$$[x]_U = A[x]_V \qquad \text{for any } \mathbf{x} \in \mathbb{R}^3$$

We know

$$[\mathbf{u}_1 \quad \mathbf{u}_2 \quad \mathbf{u}_3][x]_U = [\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3][x]_V$$

Then

$$[\mathbf{u}_1 \quad \mathbf{u}_2 \quad \mathbf{u}_3]A = [\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3]$$

Perform

$$[U|V] \to [I|A]$$

$$\begin{bmatrix} 1 & 1 & 2 & 4 & 0 & 0 \\ 1 & 2 & 3 & 6 & 1 & 1 \\ 1 & 2 & 4 & 7 & 1 & 2 \end{bmatrix} \xrightarrow[R_3 \Rightarrow R_3 - R_1]{R_2 \Rightarrow R_2 - R_1} \begin{bmatrix} 1 & 1 & 2 & 4 & 0 & 0 \\ 0 & 1 & 1 & 2 & 1 & 1 \\ 0 & 1 & 2 & 3 & 1 & 2 \end{bmatrix}$$

$$\xrightarrow[R_3 \Rightarrow R_3 - R_2]{R_1 \Rightarrow R_1 - R_2} \begin{bmatrix} 1 & 0 & 1 & 2 & -1 & -1 \\ 0 & 1 & 1 & 2 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix} \xrightarrow[R_2 \Rightarrow R_2 - R_3]{R_1 \Rightarrow R_1 - R_3} \begin{bmatrix} 1 & 0 & 0 & 1 & -1 & -2 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & -1 & -2 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

(b)From the question,

$$[x]_V = \begin{bmatrix} 2 \\ 3 \\ -4 \end{bmatrix}$$

So,

$$[x]_U = A[x]_V = \begin{bmatrix} 1 & -1 & -2 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ -4 \end{bmatrix} = \begin{bmatrix} 7 \\ 5 \\ -2 \end{bmatrix}$$

Question 8. Let $B = \left\{ \begin{bmatrix} 7 \\ 5 \end{bmatrix}, \begin{bmatrix} -3 \\ -1 \end{bmatrix} \right\}$, $C = \left\{ \begin{bmatrix} 1 \\ -5 \end{bmatrix}, \begin{bmatrix} -2 \\ 2 \end{bmatrix} \right\}$ be bases for $\mathbb{R}^2$. Find the transition matrix from B to C and the transition matrix from C to B.

Solution

Let P_{BC} and P_{CB} be transition matrix from B to C and C to B respectively, then

$$P_{BC} = C^{-1}B = \begin{bmatrix} 1 & -2 \\ -5 & 2 \end{bmatrix}^{-1} \begin{bmatrix} 7 & -3 \\ 5 & -1 \end{bmatrix} = \begin{bmatrix} -1/4 & -1/4 \\ -5/8 & -1/8 \end{bmatrix} \begin{bmatrix} 7 & -3 \\ 5 & -1 \end{bmatrix} = \begin{bmatrix} -3 & 1 \\ -5 & 2 \end{bmatrix}$$

and $P_{CB} = P_{BC}^{-1} = \begin{bmatrix} -3 & 1 \\ -5 & 2 \end{bmatrix}^{-1} = \begin{bmatrix} -2 & 1 \\ -5 & 3 \end{bmatrix}$

Question 9. Let $\mathbf{b}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \mathbf{b}_2 = \begin{bmatrix} -3 \\ 4 \\ 0 \end{bmatrix}, \mathbf{b}_3 = \begin{bmatrix} 3 \\ -6 \\ 3 \end{bmatrix}$.

(a) Show that the set $\mathbf{B} = \{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}$ is a basis of $\mathbb{R}^3$.

(b) Given a vector $\mathbf{x} = \begin{bmatrix} -8 \\ 2 \\ 3 \end{bmatrix}$ in $\mathbb{R}^3$. Find the coordinate vector of $\mathbf{x}$ with respect to the basis $\mathbf{B}$.

(c) Given the coordinate of $\mathbf{y}$ with respect to the basis $\mathbf{B}$ is $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$. What is the vector $\mathbf{y}$ in $\mathbb{R}^3$.

Solution

(a) We should prove the $\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3$ are linearly independent by row echelon form.

$$[\mathbf{b}_1 \quad \mathbf{b}_2 \quad \mathbf{b}_3] = \begin{bmatrix} 1 & -3 & 3 \\ 0 & 4 & -6 \\ 0 & 0 & 3 \end{bmatrix}$$

The row echelon form has a leading entry in each row. So $\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3$ are linearly independent and they span the space $\mathbb{R}^3$. Hence, $\mathbf{B} = \{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}$ is a basis of $\mathbb{R}^3$.

(b) $c_1\mathbf{b}_1 + c_2\mathbf{b}_2 + c_3\mathbf{b}_3 = \mathbf{x}$. Apply elementary row operations on the augmented matrix $[B|x]$.

$$\begin{bmatrix} 1 & -3 & 3 & -8 \\ 0 & 4 & -6 & 2 \\ 0 & 0 & 3 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -3 & 3 & -8 \\ 0 & 1 & -3/2 & 1/2 \\ 0 & 0 & 1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & -5 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

Hence,

$$[x]_B = \begin{bmatrix} -5 \\ 2 \\ 1 \end{bmatrix}$$

(c) $\mathbf{y} = c_1\mathbf{b}_1 + c_2\mathbf{b}_2 + c_3\mathbf{b}_3$. The scalars c_1, c_2 and c_3 are coordinates of y with respect to B, then $[y]_B = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$. So we can get: $y = 1\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + 2\begin{bmatrix} -3 \\ 4 \\ 0 \end{bmatrix} + 3\begin{bmatrix} 3 \\ -6 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ -10 \\ 9 \end{bmatrix}$

Question 10. Let U and V be subspaces of a vector space W.

Define $U + V = \{\mathbf{z}|\mathbf{z}{=}\mathbf{u}{+}\mathbf{v}$ *where* $\mathbf{u} \in U$ *and* $\mathbf{v} \in V\}$. Show that $U + V$ is a subspace of W.

Solution

Since U and V are subspaces of W, then $\mathbf{0} = \mathbf{0} + \mathbf{0} \in U + V$.

Let $\mathbf{z}_1 = \mathbf{u}_1 + \mathbf{v}_1, \mathbf{z}_2 = \mathbf{u}_2 + \mathbf{v}_2, c \in R$, where $\mathbf{u}_1, \mathbf{u}_2 \in U, \mathbf{v}_1, \mathbf{v}_2 \in V$.

So $\mathbf{z}_1 + \mathbf{z}_2 = (\mathbf{u}_1 + \mathbf{v}_1) + (\mathbf{u}_2 + \mathbf{v}_2) = (\mathbf{u}_1 + \mathbf{u}_2) + (\mathbf{v}_1 + \mathbf{v}_2)$.

Since $\mathbf{u}_1 + \mathbf{u}_2 \in U, \mathbf{v}_1 + \mathbf{v}_2 \in V$, then $\mathbf{z}_1 + \mathbf{z}_2 \in U + V$.

Also, $c\mathbf{z} = c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v} \in U + V$.

So, $U + V$ is a subspace of W.

Question 11. In $\mathbb{R}^4$, let U be the subspace of all vectors of the form $(u_1, u_2, 0, 0)^T$, and let V be the subspace of all vectors of the form $(0, v_2, v_3, 0)^T$. What are the dimensions of $U, V, U \cap V, U + V$? Find a basis for each of these four subspaces.

Solution

Since $(u_1, u_2, 0, 0)^T = u_1 (1, 0, 0, 0)^T + u_1 (0, 1, 0, 0)^T$, dim $U = 2$.
Similarly, $(0, v_2, v_3, 0)^T = v_1 (0, 1, 0, 0)^T + v_1 (0, 0, 1, 0)^T$, dim $V = 2$.
$U \cap V = (0, x, 0, 0)^T = x (0, 1, 0, 0)^T$,dim $(U \cap V) = 1$.
$U + V = (x, y, z, 0)^T = x (1, 0, 0, 0)^T + y (0, 1, 0, 0)^T + z (0, 0, 1, 0)^T$,
dim $(U + V) = 3$.

Question 12. Show that if U and V are subspaces of $\mathbb{R}^n$ and $U \cap V = \{\mathbf{0}\}$, then dim $(U + V) =$ dim $U+$ dim V .

Solution

If either $U = \{\mathbf{0}\}$ or $V = \{\mathbf{0}\}$, the result is obvious. So assume that both subspaces are nontrivial with dim $U = k > 0$ and dim $V = r > 0$.
Let $\{\mathbf{u}_1, ..., \mathbf{u}_k\}$ be a basis for U and let $\{\mathbf{v}_1, ..., \mathbf{v}_r\}$ be a basis for V. The vectors $\{\mathbf{u}_1, ..., \mathbf{u}_k, \mathbf{v}_1, ..., \mathbf{v}_r\}$ span $U + V$.
We claim that these vectors form a basis for $U+V$, hence, dim $(U+V) = k+r$.
Let $c_1\mathbf{u}_1 + ... + c_k\mathbf{u}_k + c_{k+1}\mathbf{v}_1 + ... + c_{k+r}\mathbf{v}_r = \mathbf{0}$.
Set $\mathbf{u} = c_1\mathbf{u}_1 + ... + c_k\mathbf{u}_k$ and $\mathbf{v} = c_{k+1}\mathbf{v}_1 + ... + c_{k+r}\mathbf{v}_r$, then $\mathbf{u} + \mathbf{v} = \mathbf{0}$.
Hence, $\mathbf{u} = -\mathbf{v}$, $\mathbf{u} \in U \cap V = \{\mathbf{0}\}$, so as $\mathbf{v}$.
Thus we must have $\mathbf{u} = c_1\mathbf{u}_1 + ... + c_k\mathbf{u}_k = \mathbf{0}$.
By the independence of $\{\mathbf{u}_1, \mathbf{u}_2...\mathbf{u}_k\}$, it follows that $c_1 = c_2 = ... = c_k = 0$.
Similarly, $c_{k+1} = c_{k+2} = ... = c_{k+r} = 0$.
So, $\mathbf{u}_1, \mathbf{u}_2...\mathbf{u}_k, \mathbf{v}_1, \mathbf{v}_2...\mathbf{v}_r$ are linearly independent and form a basis for $U+V$, dim $(U + V) = k + r =$ dim $U+$ dim V.